

# Transformations of Random Variables

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## Introduction

Given a random variable  $X$  and a function  $g$ , the distribution of  $Y = g(X)$  is determined by the distribution of  $X$  and the function. The change-of-variables formula computes the new distribution; for strictly monotone differentiable  $g$  this involves a Jacobian factor.

## Prerequisites

Random variables, PDF, basic calculus.

## Theory

**Univariate, monotone case.** Let  $X$  have density  $f_X$  on support  $\mathcal{X}$ , and let  $g$  be strictly monotone and differentiable on  $\mathcal{X}$ . Then  $Y = g(X)$  has density

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|.$$

The absolute value of the derivative of the inverse is the **Jacobian**.

**Non-monotone case.** If  $g$  is piecewise monotone, partition the domain into monotone segments and sum the contributions:

$$f_Y(y) = \sum_i f_X(x_i) / |g'(x_i)|$$

where  $\{x_i\}$  are the solutions of  $g(x) = y$ .

**Multivariate case.** For  $\mathbf{Y} = g(\mathbf{X})$  with  $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$  a diffeomorphism,

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(g^{-1}(\mathbf{y})) |\det J_{g^{-1}}(\mathbf{y})|,$$

where  $J_{g^{-1}}$  is the Jacobian matrix of  $g^{-1}$ .

**Common examples:**

- $Y = aX + b$ :  $f_Y(y) = f_X((y - b)/a) / |a|$ .
- $Y = X^2$ ,  $X$  standard normal:  $Y \sim \chi_1^2$ .
- $Y = e^X$ ,  $X$  normal:  $Y$  is log-normal.
- $Y = F^{-1}(U)$ ,  $U \sim \text{Uniform}(0, 1)$ : the inverse-CDF method for generating arbitrary distributions.

## Assumptions

- $g$  must be measurable.
- For the change-of-variables formula to apply,  $g$  must be strictly monotone and differentiable (or piecewise so).

## R Implementation

```
set.seed(2026)

# Case 1:  $Y = X^2$  for  $X \sim N(0, 1)$ ;  $Y$  should be chi-squared with 1 df
X <- rnorm(1e5)
Y <- X^2

par(mfrow = c(1, 2))
hist(Y, breaks = 100, freq = FALSE, main = "Empirical", col = "#2A9D8F")
curve(dchisq(x, df = 1), from = 0, to = 10, add = TRUE, col = "#F4A261", lwd = 2)

# Case 2:  $Y = \exp(X)$  for  $X \sim N(\mu, \sigma)$  is log-normal
mu <- 1; sigma <- 0.5
X2 <- rnorm(1e5, mu, sigma)
Y2 <- exp(X2)
curve(dlnorm(x, mu, sigma), from = 0, to = 10,
      main = "exp(N(1, 0.25)) vs lognormal PDF",
      col = "#2A9D8F", lwd = 2)
hist(Y2, breaks = 100, freq = FALSE, add = TRUE, border = "#F4A261")

par(mfrow = c(1, 1))

# Case 3: Inverse-CDF method to generate exponential from uniform
U <- runif(1e5)
E <- -log(1 - U) #  $F^{-1}(u) = -\log(1 - u)$  for rate = 1
c(mean = mean(E), theoretical = 1)
```

## Output & Results

| mean  | theoretical |
|-------|-------------|
| 0.999 | 1.000       |

Empirical histograms match the theoretical transformed densities. The inverse-CDF method recovers the exponential mean.

## Interpretation

Many “derived” distributions in statistics (chi-squared from normals, log-normal from normals, Cauchy from normals, Beta from gammas) arise as transformations of simpler distributions. Recognising the transformation short-cuts many derivations.

## Practical Tips

- Always check strict monotonicity before applying the simple Jacobian formula; square-mapping  $X^2$  is a classic non-monotone case.
- For inverse-CDF simulation, you need the CDF’s inverse; use `qnorm`, `qgamma`, etc., in R.
- The Jacobian for a multivariate linear transformation  $Y = AX$  is  $|\det A|$ , independent of location.
- Delta method is the “linearised” version of the change-of-variables formula, used for asymptotic distributions.
- Non-bijective transformations (e.g.,  $|X|$ ) require summing over preimages, as in the non-monotone case.

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots